MANUFACTURING CANON

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Abstract

Example

Contents

Manufacturing operations MUST be structured according to ISA-95 hierarchy with governed interfaces between enterprise and control levels.

Example: ISA-95 (IEC 62264) defines five levels of enterprise-control integration: Level 0 (physical process sensors, actuators, field devices), Level 1 (basic control PLCs, PID loops, interlocks, cycle time 100ms-1s), Level 2 (supervisory control SCADA, HMI, batch control, cycle time seconds-minutes), Level 3 (manufacturing operations management MES/MOM, scheduling, quality, maintenance, cycle time minutes-hours-days), Level 4 (business planning ERP, supply chain, CRM, cycle time days-months). Process types: discrete (countable units automotive assembly, electronics), continuous (flowing materials refining, chemicals, pharmaceuticals), batch (finite quantities food, beverage, pharmaceuticals, per ISA-88/IEC 61512). B2MML (Business to Manufacturing Markup Language) provides XML schemas for ISA-95 data exchange: production schedule, production performance, material definition, equipment information. Activity models: production (order processing, dispatching, tracking), quality (SPC, inspection, CAPA), maintenance (preventive, predictive, reactive), inventory (material tracking, WIP). MAGIC gate: bitwise AND of required governance dimensions before production order executes.

0.1 2. Industrial Cybersecurity

Industrial automation and control systems MUST be secured according to IEC 62443 with defense-in-depth across zones and conduits.

Example: IEC 62443 (Industrial Automation and Control Systems Security) series: Part 1-1 (concepts/models), Part 2-1 (security management system), Part 2-4 (service provider requirements), Part 3-2 (security risk assessment zones and conduits), Part 3-3 (system security requirements SL), Part 4-1 (secure development lifecycle SDL), Part 4-2 (component security requirements). Security Levels (SL): SL

1 (prevent casual/unintentional violation), SL 2 (prevent intentional violation using simple means low motivation), SL 3 (prevent intentional violation using sophisticated means moderate motivation, IACS-specific skills), SL 4 (prevent intentional violation using sophisticated means high motivation, state-sponsored). Zones: logical grouping of assets sharing common security requirements. Conduits: communication channels between zones each conduit has assigned SL. Reference architecture: Enterprise Zone (IT) DMZ Manufacturing Zone (Level 3) Cell/Area Zone (Level 2) Field Zone (Level 0-1). Foundational Requirements (FR): identification/authentication, use control, system integrity, data confidentiality, restricted data flow, timely response to events, resource availability.

0.2 3. C5 Technology

Industrial control systems MUST be programmed and deployed using standardized automation frameworks with verified logic.

Example: IEC 61131-3 (Programmable Controllers Programming Languages) defines five standardized PLC programming languages: Ladder Diagram (LD relay logic), Function Block Diagram (FBD dataflow), Structured Text (ST high-level procedural), Instruction List (IL assembly-like, deprecated), Sequential Function Chart (SFC state machine). Program Organization Units (POUs): functions (no state), function blocks (stateful), programs (top-level). IEC 61499 (Function Blocks for Distributed Industrial-Process Measurement and Control Systems): event-driven execution model for distributed automation basic FB (algorithms triggered by events), composite FB (networks of FBs), service interface FB (hardware/communication interface). SCADA (Supervisory Control and Data Acquisition): centralized monitoring of distributed field devices via RTUs/PLCs. Communication protocols: Modbus TCP/RTU, PROFINET, EtherNet/IP, EtherCAT, POWERLINK, OPC UA (IEC 62541). Safety PLC programming per IEC 61508 / IEC 62061 / ISO 13849: safety function blocks, diagnostic coverage, proof test intervals. SIL-capable controllers: Siemens F-CPU, Allen-Bradley GuardLogix, ABB AC500-S.

0.3 4. Industry 4.0

Smart manufacturing systems MUST implement digital integration with governed data flows across the RAMI 4.0 architecture.

Example: RAMI 4.0 (Reference Architecture Model Industrie 4.0) defines three axes: Hierarchy Levels (ISA-95 levels field device to connected world), Life Cycle & Value Stream (IEC 62890 type vs. instance), Layers (business, functional, information, communication, integration, asset). Digital Twin: virtual representation of physical asset geometry (CAD), physics (simulation), data (real-time sensor feeds), behavior (control logic). Types: digital model (manual sync), digital shadow (automatic data flow, physicaldigital), digital twin (bidirectional automatic data flow). IIoT (Industrial Internet of Things): sensor networks, edge computing (data processing at source), fog computing (intermediate layer), cloud analytics. Protocols: OPC UA (IEC 62541 platform-independent, secure, information modeling with companion specifications per industry), MQTT (ISO/IEC 20922 lightweight pub/sub messaging, QoS levels 0-2), AMQP, DDS. Edge computing: latency-sensitive processing (vibration analysis, vision inspection) at the machine level sub-millisecond response. Time-Sensitive Networking (TSN IEEE 802.1): deterministic Ethernet for converged IT/OT networks. Asset Administration Shell (AAS IEC 63278): standardized digital representation of Industry 4.0 components.

0.4 5. Supply Chain Integration

Manufacturing supply chain data exchange MUST follow standardized interfaces with governed material and production traceability.

Example: ISA-95 B2MML (Business to Manufacturing Markup Language): XML schemas for production schedule exchange between ERP (Level 4) and MES (Level 3) production request, production response, production performance. OAGIS (Open Applications Group Integration Specification): canonical business object documents (BODs) SyncProductionOrder, ProcessProductionOrder, AcknowledgeShipment. EDI (Electronic Data Interchange): ANSI X12 (US) and EDIFACT (international) standards 850 (Purchase Order), 856 (Advance Ship Notice), 810 (Invoice), 862 (Shipping Schedule). MES/MOM (Manufacturing Execution/Operations Management) per ISA-95 Part 3: production management, quality management, inventory management, maintenance management. Traceability: lot/batch tracking (FDA 21 CFR Part 11 for pharma), serialization (DSCSA for pharmaceuticals, EU FMD), genealogy (component-to-finished-good). Material flow: raw material receiving incoming inspection WIP staging production in-process inspection finished goods shipping. Kanban/JIT: pull-based material replenishment with governed minimum/maximum levels.

0.5 6. Worker Safety

Manufacturing workplaces MUST comply with occupational safety regulations with documented hazard controls and worker protections.

Example: OSHA (Occupational Safety and Health Administration) 29 CFR 1910 (General Industry Standards): Subpart D (walking/working surfaces), Subpart G (occupational health PELs for airborne contaminants), Subpart I (PPE head, eye, face, foot, hand, respiratory), Subpart J (general environmental controls LOTO per 1910.147), Subpart L (fire protection), Subpart O (machinery and machine guarding per 1910.212), Subpart S (electrical NFPA 70E arc flash). LOTO (Lockout/Tagout 29 CFR 1910.147): energy control procedures for servicing/maintenance of machines identify energy sources (electrical, mechanical, hydraulic, pneumatic, thermal, chemical, gravity), apply lockout devices, verify zero energy state, perform work, remove devices. Machine guarding per 1910.212: point of operation guards, fixed barriers, interlocked guards (per ISO 14119), light curtains (Type 2/Type 4 per IEC 61496), safety mats, two-hand controls. Confined space entry (29 CFR 1910.146): permit-required confined spaces atmospheric testing (O2, LEL, H2S, CO), ventilation, attendant, rescue plan. Hierarchy of controls: elimination substitution engineering controls administrative controls PPE.

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1. Constraints

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MUST:     Cite IEC 62443, ISA-95, OSHA, or domain-speci
MUST:     Map Security Level to MAGIC checkset governa
MUST:     Distinguish between discrete, process, and b
MUST NOT: Present ungoverned OT systems as acceptable a
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